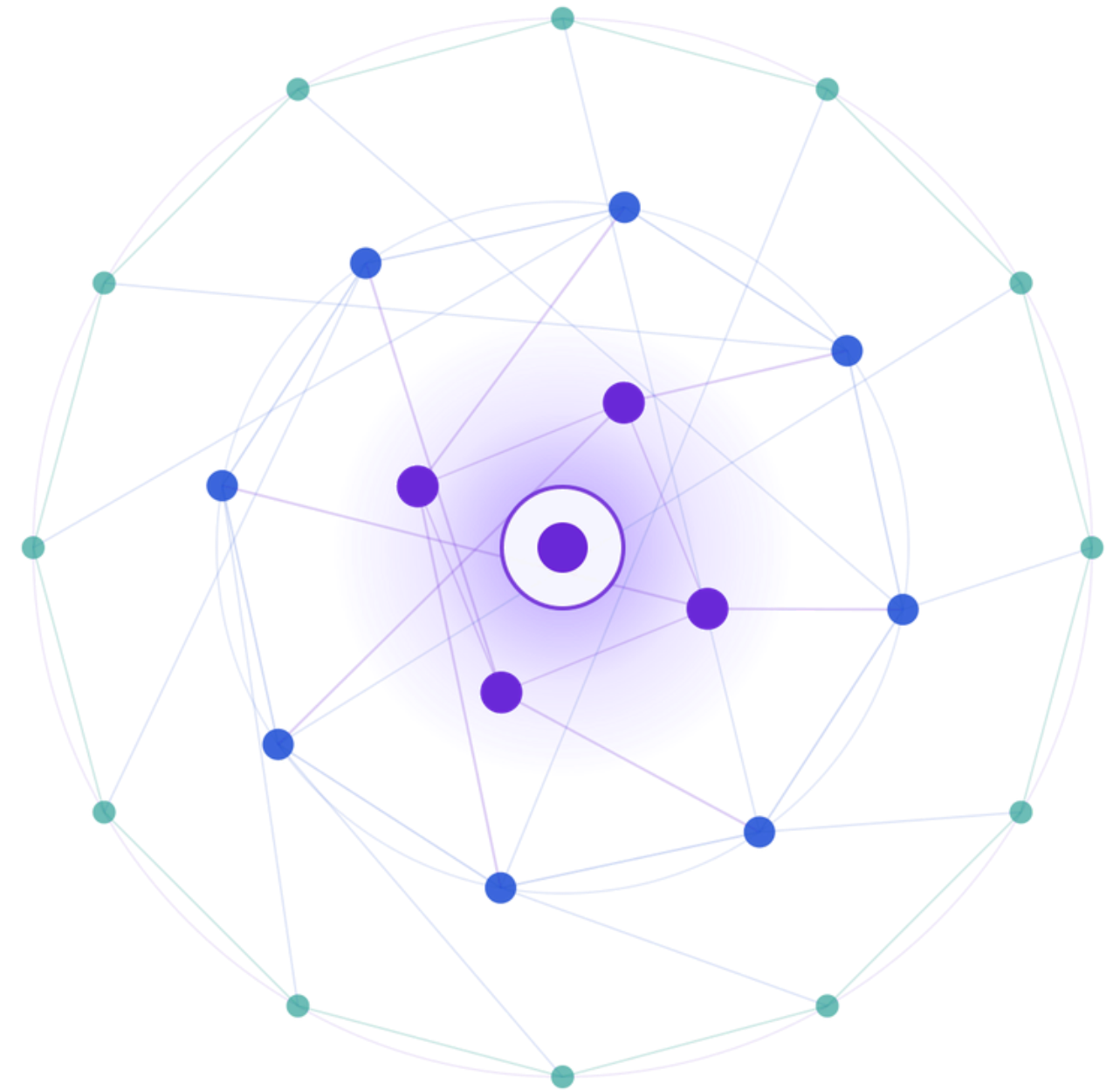


SAMSUNG PRISM

Generative AI Hackathon

3rd Edition 2026 – 27

- Theme ID - Theme 02: Smart Guided Troubleshooting Engine
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- Submission Github link - <https://github.com/tripthyganesh1797-ctrl/GENAI-HACKATHON>



Theme

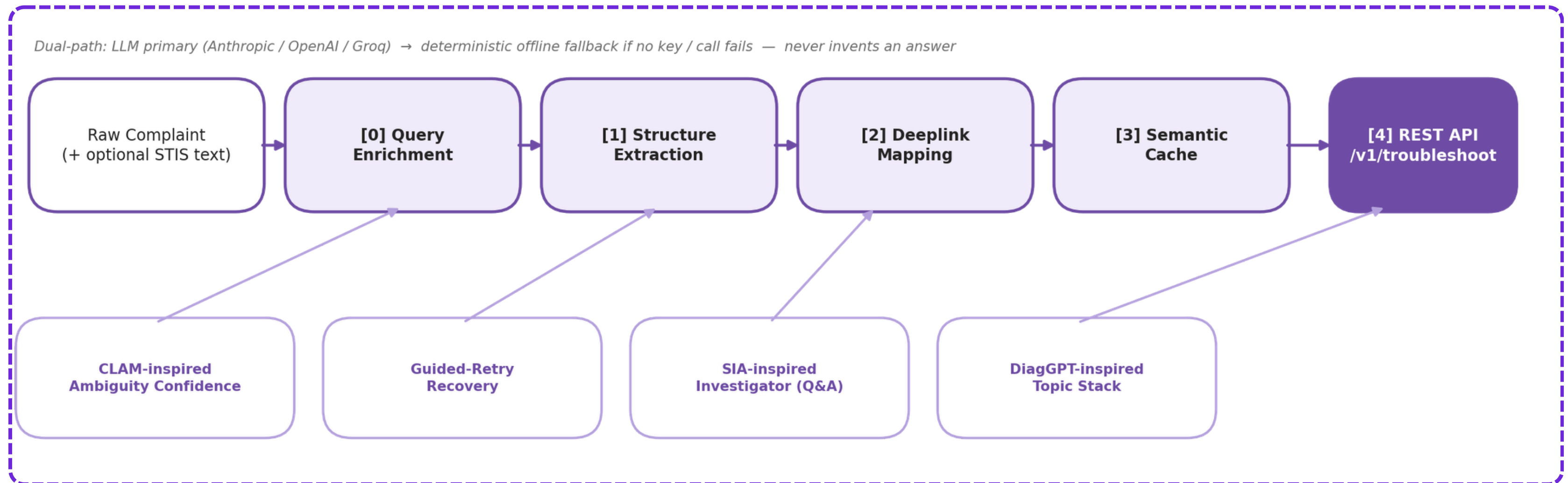
- Customers describe device problems in vague, everyday language — “screen flickers and the battery dies fast” — not standard technical terms.
- Today, resolving these complaints needs a human support agent to read unstructured knowledge-base articles, diagnose the root cause, and manually walk the customer through nested Settings menus.
- **Cost today:** This manual triage takes roughly 15 minutes per scenario, repeated across millions of customer interactions globally.
- **Target:** Build a Smart Guided Troubleshooting Engine that turns unstructured complaints into clean, validated, one-tap troubleshooting plans — with exact, verified in-app Settings deeplinks — returned in under 300ms for previously seen issues.

Existing Solutions & Gaps

- **Existing approaches:** Generic chatbots and static FAQ search return unstructured text — not a tap-able, verified action.
- Human agents reading SIIIS knowledge articles — accurate, but slow (~15 min) and inconsistent across agents.
- **Gap 1 — Hallucination risk:** A generic LLM answer can hallucinate steps or URLs that don't exist in the real product — nothing grounds it to the actual deeplink catalog.
- **Gap 2 — No ambiguity handling:** No confidence signal for when a complaint is too ambiguous to safely act on — the system just guesses.
- **Gap 3 — No failure recovery:** A single failed backend/LLM call silently drops to a worse answer with zero recovery attempt or visibility.
- **Gap 4 — No interactive diagnosis:** When 2–3 causes are equally plausible, there's no way to ask a targeted follow-up question to narrow it down.
- **Gap 5 — No session memory:** Every message is treated as a fresh, unrelated conversation, even mid-troubleshooting on the same issue.

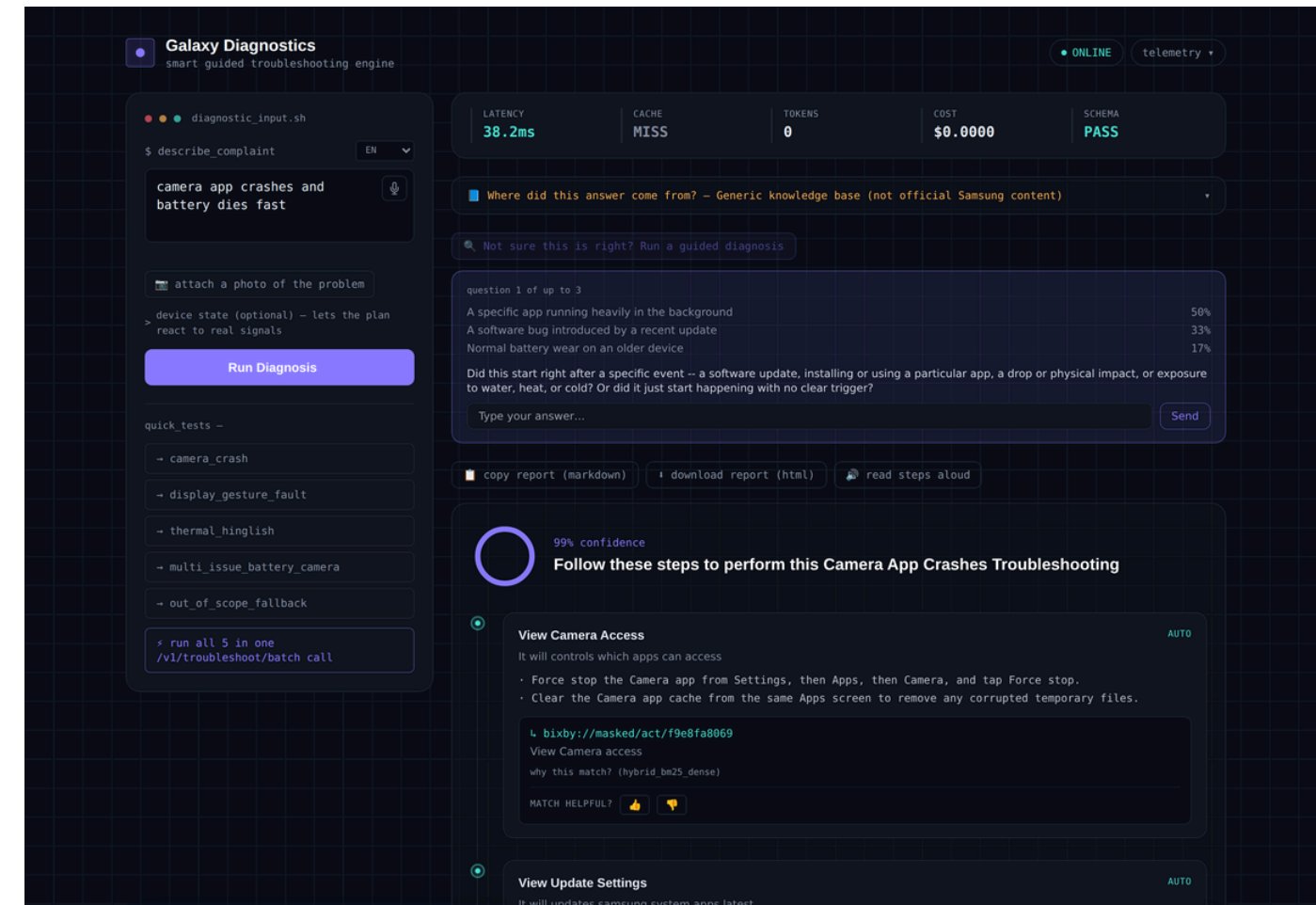
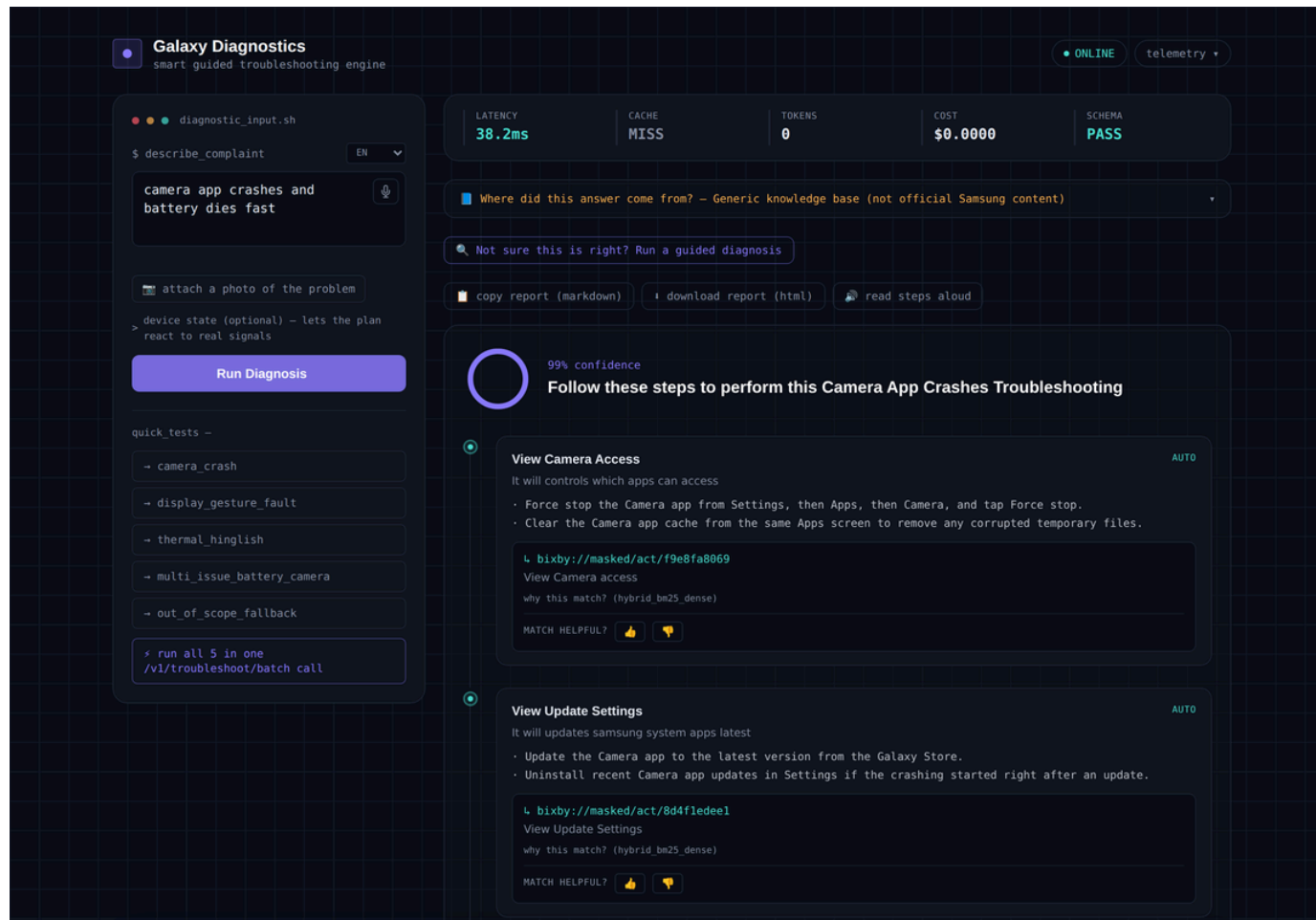
Our Solutions & Architecture Diagram

- A 5-stage dual-path pipeline — an LLM does the reasoning when a key is configured; a fully deterministic offline mode takes over automatically otherwise. Either way, it only ever returns steps and deeplinks that exist in the real catalog — honest degrade, never invents an answer.



Demo & Product Walkthrough

- Complaint in → Diagnosis result (source-transparency + deeplinked plan) → optional “Run a guided diagnosis” Q&A when confidence is low → one-tap “Did this fix it?” feedback loop.



Tools and tech stack used

PY Backend

Python 3, FastAPI, Pydantic, Uvicorn — structured errors, request IDs, per-route rate limiting.

AI LLM providers (dual-path)

Anthropic Claude / OpenAI / Groq — pluggable via env var, swappable with zero code changes.

IR Retrieval

Hybrid BM25 + dense embeddings (sentence-transformers), offline TF-IDF/SVD fallback with no network.

CI Testing & CI

pytest — 536 automated tests, ~13s offline; GitHub Actions CI; Docker.

UI Frontend & QA

Vanilla JS + Tailwind demo UI; Web Speech API voice I/O; Playwright end-to-end verification.

578 Data

578 real masked Samsung Settings deeplinks + 20 official SIIS support queries (Theme 2 dataset).

Impact & Use case

15 min → <300ms

Manual triage time per complaint, collapsed to a cache-hit response

\$0

Cost per query on the fully offline fallback tier

100%

Actions are verified catalog deeplinks — never a fabricated URL

Use case: first-line Galaxy device support deflection — a customer gets an immediate, verified, one-tap fix instead of a wall of text, and human agents only see the complaints that genuinely need a person.

Innovation highlights, results and limitations

- **Innovation:** 4 published research papers implemented directly in code — CLAM (ambiguity confidence), “When the Database Fails” (guided-retry recovery), LLM-as-an-Investigator/SIA (multi-turn probabilistic diagnosis), DiagGPT (session topic stack).
- **Results:** 100% schema compliance, 0 URL leaks, 100% deeplink catalog validity on all 20/20 official queries; P50 9.1ms / P95 60.6ms on cache hits; 100% semantic cache hit rate on unseen paraphrases; 536/536 tests passing offline.
- **Known limitations (stated honestly):** Offline relevance gating is a bag-of-words heuristic (the LLM path is more robust); dense retrieval falls back to TF-IDF/SVD without HuggingFace Hub access; Hindi/Hinglish support is a ~135-word phrasebook, not full translation; investigator/session state is in-process, not persisted.

Results & Metrics

536 / 536

Automated tests passing, fully offline

100%

Schema-valid, rule-compliant output

20 / 20

Official SIIS query coverage

0

Fabricated / leaked URLs

100%

Semantic cache hit rate on paraphrases

system_telemetry — /stats · live, computed from real request/feedback logs

REQUESTS SERVED

1

CACHE HIT RATE

0%

NO-MATCH RATE

0%

P95 LATENCY

38.2ms

REQUESTS BY DOMAIN

Battery

1

DEEPLINK MATCH FEEDBACK (0 DEEPLINKS RATED)

no feedback submitted yet -- try the 👍/👎 buttons under a matched deeplink

TRENDING ISSUES -- TOP RECURRING COMPLAINTS

Battery draining fast

1

table view (all numbers, screen-reader friendly)

Live from the demo UI's /stats telemetry panel

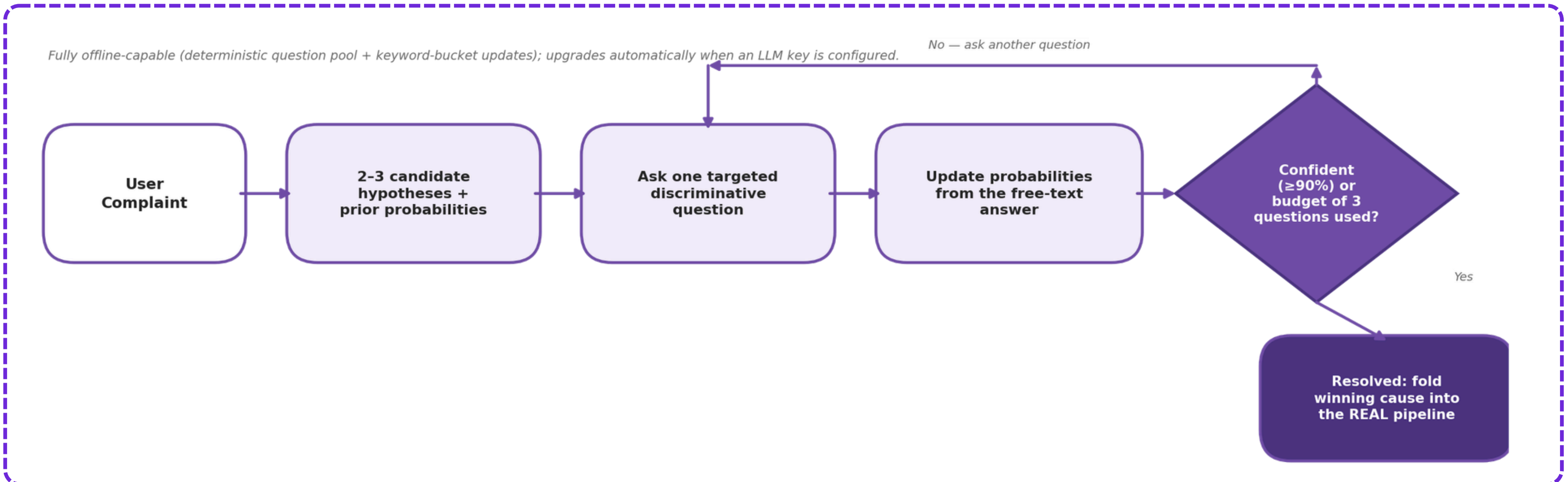
Latency (cached path)

P50: 9.1 ms • P95: 60.6 ms

Target was ≤ 300 ms — beaten by roughly 5x

How the Guided Investigator Works

SIA-inspired: instead of guessing, the engine narrows down between 2–3 plausible causes with short, targeted questions — then hands the real answer to the real pipeline, never to a made-up one.



What's next

- Switch on real sentence-transformer embeddings once deployed outside a network-restricted sandbox — the code already supports this, it just needs Hub access.
- Move investigator and topic-stack session state from in-process memory to Redis/DB-backed storage so it survives a restart.
- Grow the Hindi/Hinglish phrasebook toward fuller language coverage.
- Extend past the Screen/Display domain to the rest of the 578-entry deeplink catalog (Battery, Camera, Performance, Connectivity).
- Explore an on-device / edge deployment path for fully offline-first support.

Brownie points slide (differentiation)

4

Research, not just citations

4 published papers actually wired into working, tested code — each module's docstring documents the honest mapping between the paper's method and what's really implemented.

\$0

Offline fallback is real

The whole system runs and passes its full 536-test suite with zero API key and zero cost — not a stub.

✓

536 tests + CI

Every new feature shipped with real automated test coverage from day one, not just a demo script.

+

Beyond the spec

Voice input/output, Hindi/Hinglish support, multi-issue detection, match explainability, shareable reports, and a post-fix feedback loop.

Thank you